COMPLEX EIGENVALUES

Over $\mathbb{R}$ vs over $\mathbb{C}$

The choice of field decides how far the characteristic polynomial factors.

Over the reals $\mathbb{R}$	VS	Over the complex numbers $\mathbb{C}$
Linear factors for the real eigenvalues, and irreducible quadratic factors for the rest	CHARACTERISTIC POLYNOMIAL	Every polynomial of degree n factors into n linear factors (Fundamental Theorem of Algebra)
Real eigenvalues, plus complex conjugate pairs hidden inside the quadratic factors	EIGENVALUES	Exactly n eigenvalues, counted with algebraic multiplicity
Not every matrix is triangularizable over $\mathbb{R}$	TRIANGULAR FORM	Every matrix is triangularizable: $A = UTU^*$ with U unitary and the eigenvalues on the diagonal of T (Schur)
Diagonalizability can fail for a reason the real field adds: the irreducible quadratics	DIAGONALIZABILITY	Depends only on whether the geometric multiplicities match the algebraic ones
The natural setting for computation	ROLE	The natural setting for spectral theory

BOTH PERSPECTIVES

A real matrix is computed with over $\mathbb{R}$ and understood over $\mathbb{C}$: the same matrix, two fields.